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Visible-light-Enabled O-Selective Radical Etherification of 1,2-Dicarbonyl Compounds for the Synthesis of Sterically Congested Ethers

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Abstract

A visible-light-promoted, triphenylphosphine-mediated protocol for the direct O-selective etherification of 1,2-diketones is reported. Using NHPI esters as radical precursors, this method provides an efficient route to sterically congested ethers under mild conditions. The strategy exhibits compatibility with various tertiary alkyl radicals and 1,2-diketone derivatives. Mechanistic studies support a radical-mediated pathway, and adaptation to a photochemical continuous-flow platform enables improved efficiency, scalability, and operational control.

Keywords

visible-light-promoted, O-selective etherification, sterically congested ethers, photochemical continuous-flow

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Visible-light-Enabled O-Selective Radical Etherification of 1,2-Dicarbonyl Compounds for the Synthesis of Sterically Congested Ethers

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Keywords: visible-light-assisted photocatalysis, O-selective etherification, sterically congested ethers, continuous-flow

Abstract

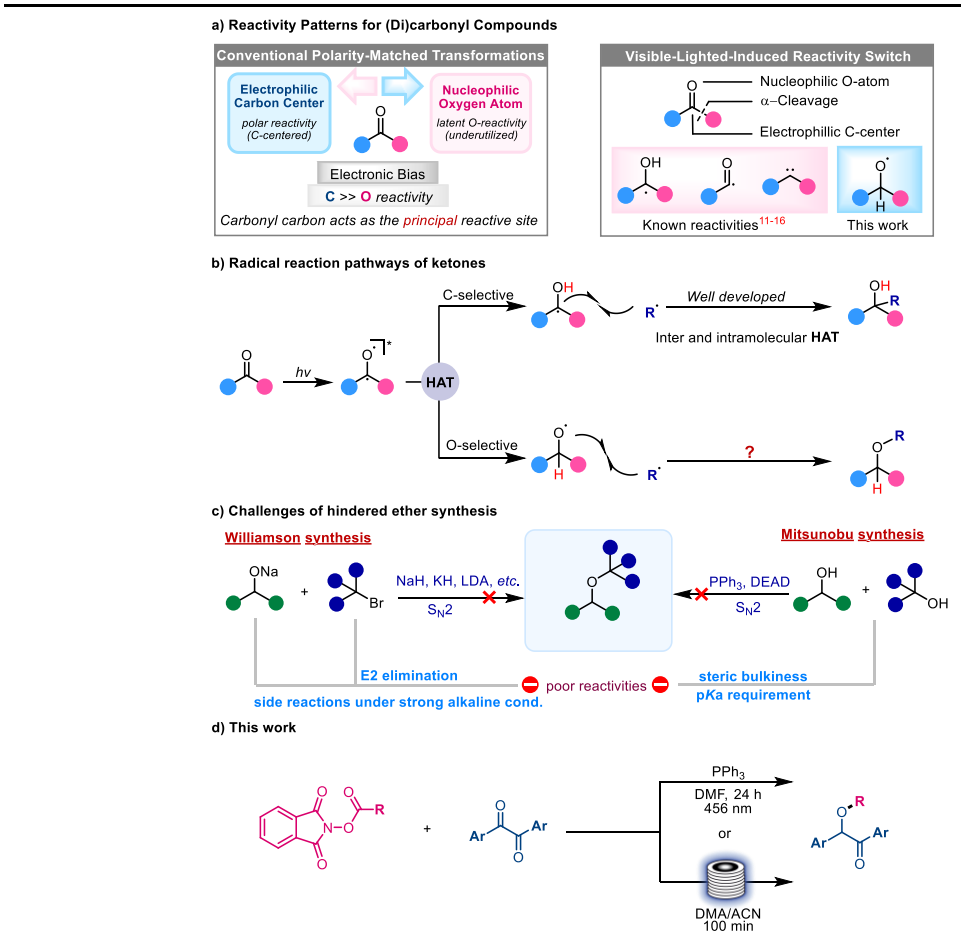
A visible-light-promoted, triphenylphosphine-mediated strategy enables the direct O-selective etherification of 1,2-diketones using NHPI esters as alkyl radical precursors. The method provides an efficient route to sterically congested ethers under mild conditions and accommodates a range of tertiary alkyl radicals and 1,2-diketone derivatives. Mechanistic experiments and DFT calculations support a radical-mediated pathway. Integration into a photochemical continuous-flow platform improves reaction efficiency and scalability while providing greater operational control.

Introduction

The carbonyl group, particularly as found in ketones, is one of the most important and versatile functional motifs in organic synthesis.[1-3] Its distinctive electronic structure, with an electrophilic carbon center and a nucleophilic oxygen atom, makes compounds containing it valuable building blocks for the construction of molecular complexity.[4-7] In most conventional transformations, the reactivity of ketones is dominated by polar processes in which the carbonyl carbon functions as the principal reactive site (**Scheme 1a**).[8-10]

In radical chemistry, this preference for carbon-centered reactivity is even more pronounced.[11-16] Over the past decade, radical-mediated functionalization of carbonyl compounds has evolved into a powerful strategy in modern synthetic chemistry.[17-18] Extensive studies have shown that carbon-centered radicals readily add to the electrophilic carbonyl carbon,[19] providing efficient access to a broad range of secondary and tertiary alcohol derivatives.[20-22] This carbon-attack mode, frequently enabled by photoredox or transition-metal catalysis, has become a reliable platform for C-C bond formation.[23-26]

Scheme 1. Challenges in hindered-ether synthesis and the concept of light-induced reactivity switch of ketones. **a)** Reactivity patterns for (di)carbonyl compounds; **b)** Radical reaction pathways of ketones; **c)** Challenges in hindered-ether synthesis; **d)** This work.



In contrast, the direct engagement of the carbonyl oxygen atom in radical processes has received comparatively little attention. Although the oxygen atom inherently possesses nucleophilic character, both kinetic and thermodynamic factors generally favor radical addition to the carbon center.[27-28] For 1,2-diketones, however, the calculations reported below show that the carbon adduct, although kinetically accessible, lies above the separated reactants and reverts, which removes the thermodynamic component of this selectivity preference. This means that, the large unexplored radical-mediated O-alkylation of carbonyl compounds, where selective interaction occurs at the oxygen atom to form C-O bonds, becomes competitive (**Scheme 1b**).

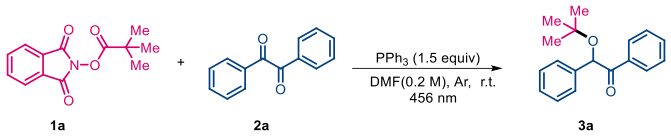
The challenge becomes particularly evident in the synthesis of sterically hindered ethers. Classical approaches, such as the Williamson ether synthesis (**Scheme 1c**),[29-32] often perform poorly with bulky substrates owing to slow S_N2 kinetics and competing elimination pathways.[33-34] Radical-mediated O-alkylation offers a mechanistically distinct alternative.[32-35] By circumventing the steric constraints associated with polar substitution reactions, this strategy holds promise for the efficient construction of highly substituted and sterically congested ether architectures that are otherwise difficult to access through traditional methods.

Herein, we report a visible-light-promoted, PPh_3-mediated protocol for the direct etherification of 1,2-diketones (**Scheme 1d**). The method enables O-selective coupling with sterically demanding substrates, thereby facilitating the formation of sterically congested ethers that are difficult to access through classical S_N2 pathways. By expanding the functionalization landscape of carbonyl oxygen atoms, this strategy provides a concise and atom-economical approach to structurally complex-ether architectures.

Our initial investigations commenced with 1,3-dioxoisoindolin-2-yl pivalate **1a** and benzil **2a** as the model substrates (Table 1 and Table S1-S5 in the supporting information). When 1.5 equivalents of triphenylphosphine were employed in DMF (0.2 M) under irradiation with blue LEDs at room temperature for 24 h, the desired sterically hindered ether product **3a** was obtained in 72% yield (Table 1, entry 1). To further refine the reaction conditions, a comprehensive screening of solvents across a broad polarity gradient was conducted, including DMA, MeCN, THF, acetone, EtOAc, and DMSO (entries 2-7). Despite these efforts, DMF remained the optimal choice, as other solvents failed to enhance the reaction efficiency. Furthermore, the impact of the light source was investigated; switching to a shorter wavelength (390 nm) resulted in a significant drop in yield to 45%, underscoring the critical role of specific visible-light irradiation in facilitating this transformation (entry 8). The origin of this wavelength dependence is addressed in the computational section ahead. The stoichiometry of triphenylphosphine was found to be crucial for the reaction efficiency, as evaluated through a range of 0.5 to 2.0 equivalents (entries 9-11). It was observed that 1.5 equivalents of PPh_3 represented the optimal balance; increasing the loading to 2.0 equivalents led to a noticeable decline in yield (60%). Conversely, diminishing the amount of PPh_3 to 1.0 or 0.5 equivalents resulted in significantly lower yields of 56% and 30%, respectively, indicating that an adequate

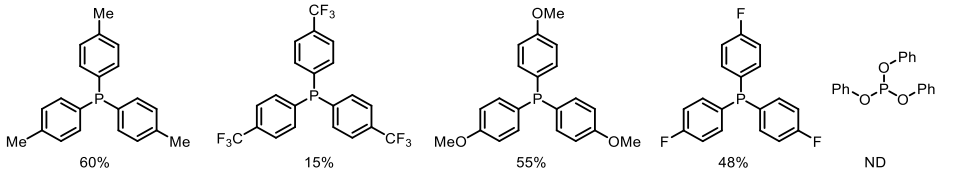
concentration of the mediator is essential to sustain the transformation. Control experiments were subsequently conducted to evaluate the necessity of each reaction component (entries 12-14). The results clearly demonstrated that the transformation did not proceed in the absence of triphenylphosphine or when kept in the dark, confirming that both the PPh_3 mediator and visible-light irradiation is indispensable for the etherification process. Furthermore, the reaction exhibited sensitivity to atmospheric conditions; performing the coupling under an ambient air atmosphere led to a significantly diminished yield of 41%, suggesting that an inert environment is required to prevent potential radical quenching or side reactions.

Table 1: Optimization of reaction conditions.[a]



1a 2a 3a

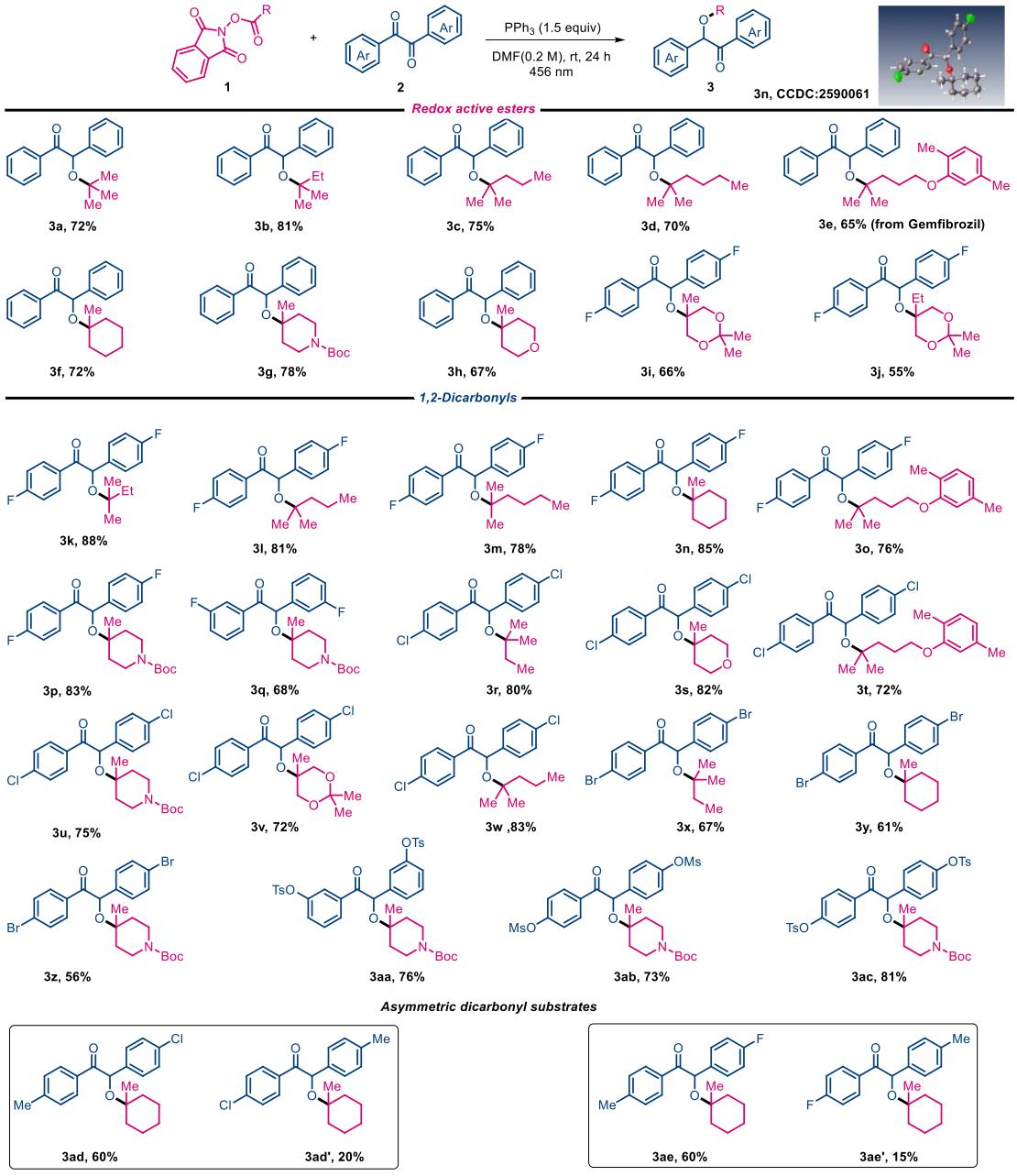
Entry	Deviation from the standard conditions	Yield (%)
1	None	72
2	DMA instead of DMF	52
3	MeCN instead of DMF	15
4	THF instead of DMF	25
5	Acetone instead of DMF	30
6	EtOAc instead of DMF	17
7	DMSO instead of DMF	Trace
8	Purple LEDs (390 nm) instead of blue LEDs	45
9	2.0 equiv PPh_3 was used	60
10	1.0 equiv PPh_3 was used	56
11	0.5 equiv PPh_3 was used	30
12	Without PPh_3	NR
13	In the dark	NR
14	under air	41



[a]Reaction conditions: **1a** (0.4 mmol), **2a** (0.2 mmol), PPh_3 (1.5 equiv), DMF (1.0 mL), irradiation at blue LEDs under an argon atmosphere for 24 h at room temperature.

With the optimized reaction conditions in hand, we first evaluated the substrate scope regarding various tertiary N-hydroxyphthalimide (NHPI) esters as radical precursors (**Scheme 2**). Under the standard conditions, both acyclic and cyclic tertiary alkyl radicals were well-tolerated, delivering the corresponding products in moderate to excellent yields. Initially, the reaction of

Scheme 2. Reaction scope investigation for the etherification reaction[a]

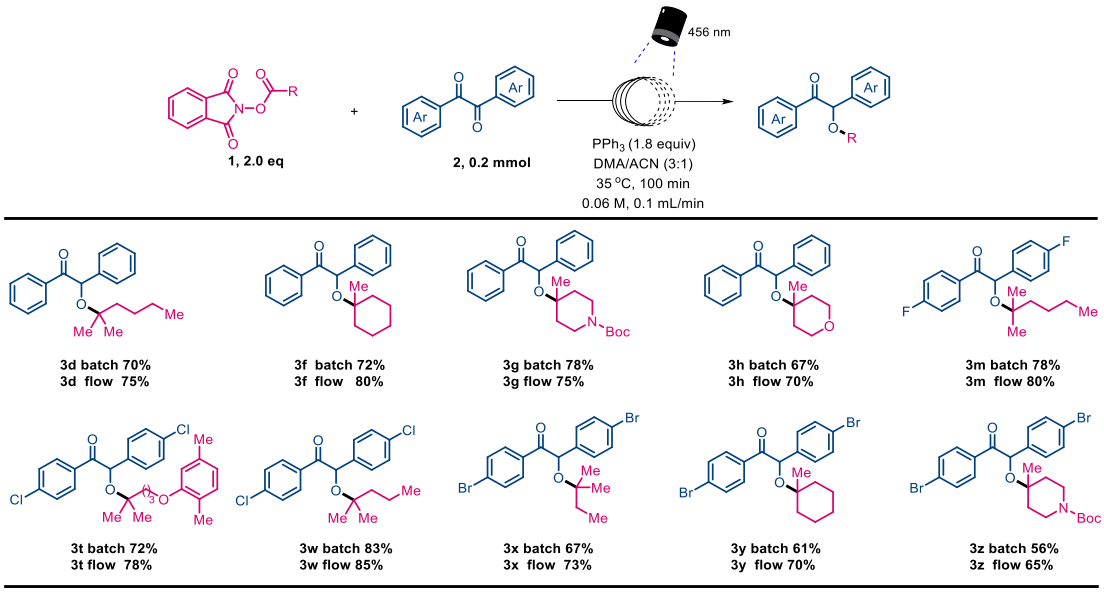


[a]**Reaction conditions**: **1** (0.4 mmol), **2** (0.2 mmol), PPh_3 (0.3 mmol), DMF (1.0 mL); irradiation at blue LEDs under Ar atmosphere for 24 h at room temperature.

NHPI esters bearing tert-butyl (**1a**) and dimethylbutyl (**1b**) groups with **2a** afforded products **3a** and **3b** in 72% and 81% yields, respectively. Substrates with longer alkyl chains, including dimethylpentyl (**1c**) and dimethylhexyl (**1d**) derivatives, also proceeded smoothly to provide the corresponding products **3c** in 75% yield and **3d** in 70% yield. Notably, the NHPI ester derived from the bioactive molecule Gemfibrozil was also compatible with this transformation, yielding product **3e** in 65% yield. The methylcyclohexyl-substituted substrate**1f** performed well, providing **3f** in 72% yield. In addition, heterocyclic motifs such as N-Boc-piperidine (**1g**) and tetrahydro-2H-pyran (**1h**) were successfully converted into their corresponding products **3g** and **3h** in 78% and 67% yields, respectively. The 1,3-dioxane substituted NHPI ester also proved to

be a viable substrate, furnishing the etherified products **3i** and **3j** in 66% and 55% yields. Simultaneously, the scope of 1,2-diketone derivatives was explored (**Scheme 2**). Using 1,2-bis(4-fluorophenyl) ethane-1,2-dione as the substrate, the reaction proceeded efficiently to furnish products **3i-3p** in excellent yields. Similarly, the meta-fluorinated analogue was well-tolerated, providing **3q** in 68% yield. The 4-chlorophenyl derivative also exhibited high reactivity, delivering products **3r–3w** in excellent yields, while the 4-bromophenyl counterpart afforded products **3x–3z** in moderate yields. Beyond halogenated substrates, 1,2-diketones containing sulfonyl groups were successful, providing products **3aa-3ac** in high yields. Notably, when asymmetrical 1,2-diketones **2h** and **2j** were employed, the reaction exhibited regioselectivity, affording **3ad/3ad′** and **3ae/3ae′** in 60:20 and 60:15 ratios, respectively.

Scheme 3. Etherification reaction under Photo-Flow conditions.[a]

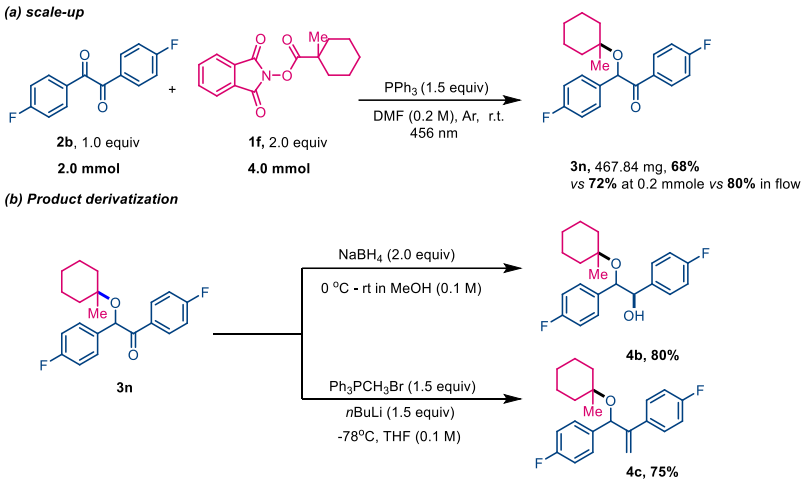


[a]Reaction conditions: **1a** (0.4 mmol), **2a** (0.2 mmol), PPh_3 (0.36 mmol), DMF/ACN (3.0 mL); at 0.1 mL/min with a reactor volume of 10 mL, residence time of 100 min, and irradiation with blue LEDs (40 W).

To demonstrate the practical utility of this protocol, the transformation was adapted to a photochemical continuous-flow reactor resulting in significant enhancement in the reaction rate with the improved product yield in most cases for a 100 min reaction as compared to traditional batch conditions (**Scheme 3**). For the detailed optimization of continuous-flow parameters such as concentration, temperature, flow rate and residence time see supporting information.

In addition, the scalability of this protocol was further evaluated by performing the reaction on a 2.0 mmol scale (**Scheme 4a**). Under these conditions, the product was obtained in 68% yield, confirming that the transformation remains efficient beyond small-scale screening. To explore the synthetic potential of the obtained ethers (**Scheme 4b**), the carbonyl group was subjected to reduction using sodium borohydride, which smoothly furnished the corresponding alcohol **4b**. In addition, an olefin **4c** was also successfully prepared *via* the Wittig reaction.

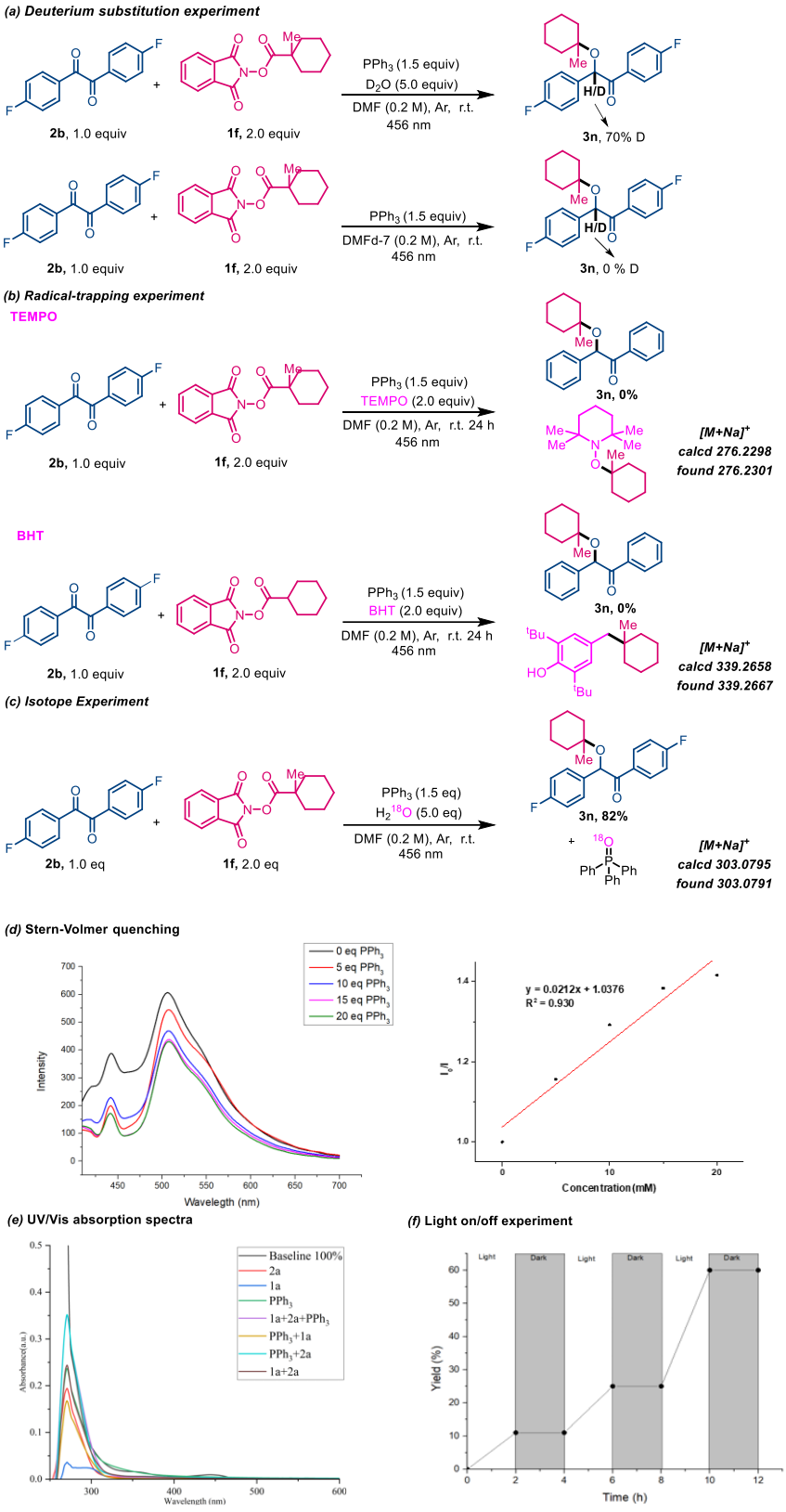
Scheme 4. Batch scale-up reaction and product derivatization.



This successful derivation highlights the value of these sterically hindered ethers as intermediates for further structural diversification.

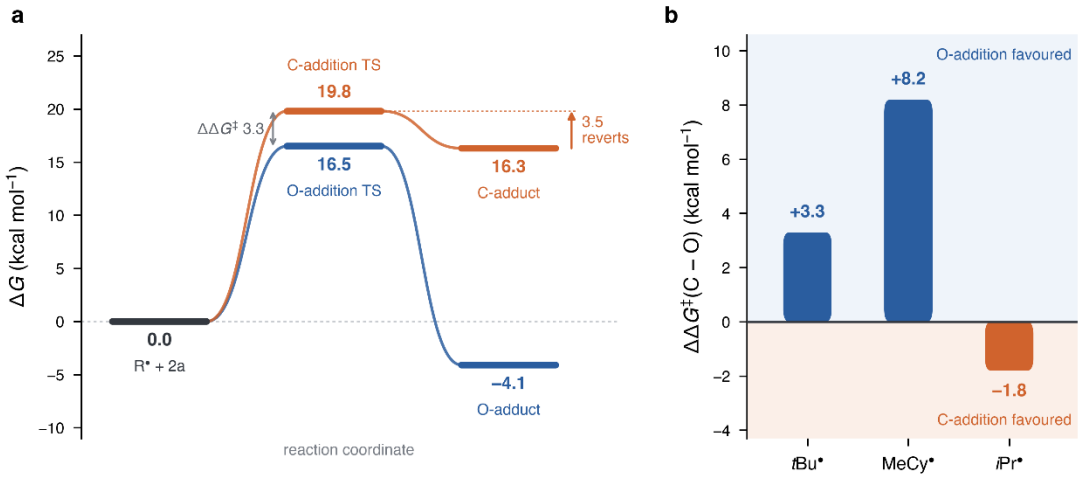
To elucidate the reaction mechanism, several control experiments were conducted: the addition of D_2O to the reaction system resulted in 70% deuterium incorporation in the final product. In contrast, when DMF-d7 was employed as the solvent, no deuterated product was observed (**Scheme 5a**). This suggests that the hydrogen atom in the product likely originates from trace amounts of water present in the solvent rather than the solvent itself (for details on the effect of water, see Table **S5** in the supporting information). The addition of 2.0 equivalents of radical scavengers, such as TEMPO or BHT, completely inhibited the reaction. Furthermore, both the TEMPO-alkyl radical and BHT-alkyl radical adducts were detected *via* HRMS analysis, supporting a radical-mediated pathway (**Scheme 5b**). Additionally, isotope-labeling experiments were conducted: when 5.0 equivalent of $H_2{}^{18}O$ was used, HRMS detected the formation of ^{18}O-labeled triphenylphosphine oxide (**Scheme 5c**). This observation is consistent with the hypothesis that H_2O acts as the oxygen source for the formation of triphenylphosphine oxide. Subsequently, Stern-Volmer fluorescence quenching experiments were performed to investigate the single electron transfer pathway between excited-state **2a** and PPh_3. As depicted in **Scheme 5d** the fluorescence intensity of a 1.0 mM **2a** solution in DMF was progressively quenched upon the incremental addition of PPh_3 under an argon atmosphere (see the supporting information for details). These results suggest that an electron transfer process likely occurs between excited state **2a** and PPh_3. Meanwhile, to explore the potential formation of EDA complexes, UV-Vis spectroscopic characterization of the reaction components was carried out (**Scheme 5e**). As shown in the figure, no significant red shift was observed among PPh_3, **1a**, and **2a**, suggesting that no detectable EDA complex was formed under the reaction conditions. Additionally, light on/off experiments demonstrated that the reaction ceased upon removal of light, indicating that continuous irradiation is essential for the reaction to proceed (**Scheme 5f**).

Scheme 5. Control experiments and mechanistic investigations.



To validate the experimental observations and gain deeper theoretical insights into the reaction pathway, several mechanistic aspects of this process were investigated using density functional calculations (see the supporting information for details).

Scheme 6. Computational study of the mechanism.



Reaction energetics for radical addition to diketone **2a**. (a) Free-energy profiles for *O*- and *C*-addition of the tert-butyl radical. (b) Comparison of addition barriers for tertiary versus secondary radicals.

The free energies were determined by DLPNO-CCSD(T)/def2-TZVP single point calculations on geometries obtained with the ωB97X-D3/def2-TZVP functional and a CPCM(DMF) reaction field, all referred to the 1 M standard state at 298.15 K.

Both of the electron transfers are possible. The oxidative quenching of the triplet state of the diketone by PPh_3 is exergonic in all three cases (it is −5.1 kcal mol^{-1} for compound **2a**, −6.6 kcal mol^{-1} for its 4,4′-difluoro derivative and −2.6 kcal mol^{-1} for the 4,4′-dimethoxy compound), and the following reduction of the NHPI ester by the radical anion thus formed is nearly thermoneutral, the sign changing depending on the substitution (+2.2, +2.7 and −2.2 kcal mol^{-1} respectively; see the supporting information for details). The Marcus activation free energies are 7.09 and 8.56 kcal mol^{-1}, both compatible with reaction under blue-light irradiation at room temperature. The fragmentation of the reduced ester to give the alkyl radical, carbon dioxide, and the phthalimide anion then takes place. In addition, the diminished thermodynamic driving force for electron-rich derivatives helps account for the lower efficiency observed with diketones bearing electron-donating groups.

The isotope-labeling result is explained by investigating phosphorus reactivity. When the triphenylphosphine radical cation undergoes hydration, the oxygen from water is transferred to the molecule to form $Ph_3P{=}O$; this is why ^{18}O-labelled phosphine oxide is detected when the reaction is carried out using $H_2{}^{18}O$. The alternative pathway involving the closed-shell hydroxy phosphorane, $Ph_3P(H)OH$, is not feasible in this process because formation of this intermediate from PPh_3 and water is endergonic by 30.4 kcal mol^{-1}, corresponding to an equilibrium concentration of about 10^{-23} M. The open-shell pathway is therefore accessible.

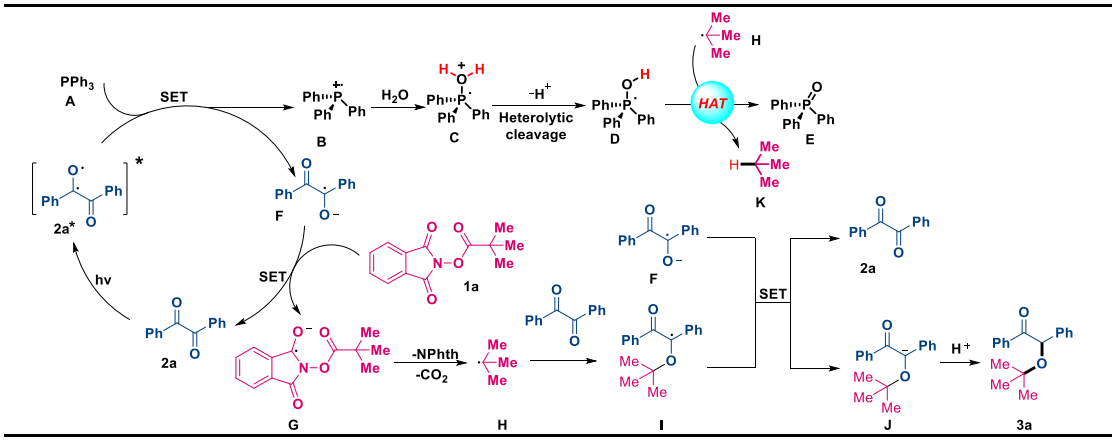
Regarding selectivity, the addition of the tert-butyl radical to the carbonyl oxygen of the diketone occurs over a free-energy barrier of 16.5 kcal mol^{-1} and is exergonic by 4.1 kcal mol^{-1}, leading to a carbon radical which is captodatively stabilized and already contains the alkyl-oxygen bond of the product (**Scheme 6a**). In contrast, addition at the carbonyl carbon (the regiochemistry that would give the alcohol) is disfavored in two ways: its transition state is 3.3 kcal mol^{-1} higher, and the resulting C-adduct is situated 16.3 kcal mol^{-1} above the separated

reactants, with a reverse barrier of only 3.5 kcal mol^{-1}, so it reverts to the starting materials rather than proceeding further. The reversal from the usual behavior of carbonyl compounds arises because the oxygen adduct benefits from captodative stabilization, whereas the carbon adduct with an alkoxy group does not.

It is this step that explains why the scope is limited to tertiary esters (as shown in **Scheme 6b**): for the tertiary 1-methylcyclohexyl radical the preference for oxygen increases to 8.2 kcal mol^{-1}, but for the secondary isopropyl radical it decreases and reverses to -1.8 kcal mol^{-1}, together with an increase in the absolute barrier for O-addition. The restriction is thus a characteristic of the bond-forming step itself and also indicates a sharp not a gradual transition; secondary esters are predicted to fail completely rather than merely giving lower yields of the ether. These findings are consistent with the experimental substrate scope, indicating a clear preference for tertiary radicals over secondary ones.

Finally, time-dependent density functional theory calculations show the vertical n→π* transitions for **2a** to occur at 3.19 eV (T_1) and 3.65 eV (S_1, with an oscillator strength approximately equal to 10^{-3}), the adiabatic triplet energy being 2.48 eV. As a result, illumination with blue light of wavelength 2.72 eV excites within the vibronically broadened onset of a weak and symmetry-disfavoured band and leads to a correspondingly low rate of triplet formation. Irradiation with shorter wavelength light is more efficient at exciting the band and thus increases that rate, a result which agrees with the reduced yield observed at 390 nm.

Scheme 7 Possible mechanism



Based on mechanistic investigations and relevant literature reports,[35–38] a plausible reaction pathway is proposed for this transformation (**Scheme 6**). Initially, photoexcitation of **2a** under light irradiation yields the excited state **2a***, which undergoes a single-electron transfer process with PPh_3 to generate triphenylphosphine radical cation **B** and radical anion intermediate **F**. Subsequently, **F** reduces the radical precursor **1a** via another SET step, regenerating **2a** alongside radical anion intermediate **G**. Extrusion of CO_2 and a phthalimide anion ($NPhth^-$) from **G** affords the alkyl radical **H**. Then, radical **H** attacks **2a** to generate the carbon-centered radical intermediate **I**. Next, intermediate **I** undergoes a single-electron transfer (SET) process with **F** to form carbanion intermediate **J** along with the regeneration of **2a**. Finally, protonation of **J** yields the desired product **3a**. Meanwhile, the phosphine radical cation **B** reacts with trace H_2O followed by deprotonation to yield the radical intermediate **D**. A subsequent hydrogen atom transfer between **D** and the tert-butyl radical **H** affords triphenylphosphine oxide **E** along with isobutane **K** as a byproduct.

In summary, we have developed a visible-light-promoted, PPh_3-mediated protocol for the direct O-selective etherification of 1,2-diketones. This method provides a practical approach to constructing sterically congested ethers that are typically challenging to access via classical S_N2 pathways. The transformation proceeds under mild conditions and exhibits compatibility with various NHPI esters and 1,2-diketone derivatives. Mechanistic investigations, supported by radical trapping experiments and DFT calculations, suggest a radical-mediated process. Furthermore, the adaptation to a photochemical continuous-flow reactor demonstrates the protocol's potential for improved reaction efficiency and synthetic utility.

Conflicts of interest

There are no conflicts to declare.

Data availability

All minima and transition states discussed in the computational section are made available in computational_structures.zip.

Acknowledgments

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